

Association of socio-demographic factors and pain presentation in a cohort of 742 chronic pain patients

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Introduction

Chronic pain is a complex and multifaceted condition that manifests in highly heterogeneous ways across individuals. Much variability is observed in the intensity of pain, the response to conservative treatment, and scores on other related patient-reported outcome measures (PROMs) such as anxiety and depression. The aim of this study was to estimate associations between socio-demographic characteristics and pain intensity in patients with chronic pain at the intake in a specialized pain clinic. The secondary aim was to estimate correlations between PROMs in patients with chronic pain.

Methods

For this study, we used a subset of patients from the chronic pain registry of the Maastricht University Medical Centre department of pain management who had been recruited between January 2022 and December 2024. This registry has been described in great detail elsewhere Brouwer et al. (2020). Patients were considered eligible if they had pain for at least 3 months and conservative treatment (e.g. physical therapy) failed. Patients received questionnaires at enrollment, consisting of several PROMs, such as the RAND-36 questionnaire and the hospital anxiety and depression scale (HADS). Baseline data were summarized as mean and standard deviation (SD) or as count and percentage. Multivariable linear regression analysis was used to estimate associations between socio-demographic factors and pain intensity. Pearson's correlation coefficient was computed to estimate the correlation between pain intensity, depressive complaints, and anxiety.

Results

Baseline characteristics

Between January 2022 and December 2024, we included 742 chronic pain patients, of whom 411 (55.4 %) were female, with a mean age (SD) of 57.18 (15.14). Table 1 shows baseline characteristics of the cohort of chronic pain patients.

Table 1: Baseline characteristics

Characteristic	Total cohort
Age (years)	57.18 (15.14)
Sex (female)	411 (55.4%)
Partner (y)	555 (74.8%)
Child (y)	591 (79.6%)
PCS score	28.08 (11.98)

Associations between socio-demographic factors and pain intensity

The mean (SD) pain intensity measured using the RAND questionnaire pain domain was 51.99 (12.97). Linear regression analysis did not reveal any statistically significant associations between socio-demographic factors and pain intensity (table 2). However, both female sex (regression coefficient = 1.077) and having a child (regression coefficient = 1.965) were associated with a clinically meaningful higher mean pain intensity, albeit not significant.

Table 2: Table 2 Association between baseline characteristics and pain intensity

	<i>Dependent variable:</i>
	Pain intensity
Sex (female)	1.077 (−0.848, 3.003) p = 0.274
Age (years)	0.047 (−0.019, 0.113) p = 0.163
Partner (yes)	−0.307 (−2.576, 1.962) p = 0.791
Child (yes)	1.965 (−0.522, 4.452) p = 0.122
Constant	46.260 (40.807, 51.712) p = 0.000***
Observations	723
R ²	0.010
Adjusted R ²	0.005
Residual Std. Error	12.977 (df = 718)
F Statistic	1.853 (df = 4; 718)
<i>Note:</i>	*p<0.1; **p<0.05; ***p<0.01

Correlations between outcome measures

Figure 1 shows the correlations between different outcome measures. Strong negative correlations were seen between the HADS depression and anxiety scales on the one hand, and the RAND questionnaire mental health dimension, and moderate positive correlations between the two HADS subscales. The RAND pain dimension was not strongly correlated to any of the other patient-reported outcomes.

Conclusion

We did not find strong significant associations between socio-demographic factors and pain intensity in a large cohort of chronic pain patients, albeit some were potentially clinically relevant. Pain intensity was not strongly correlated to other patient outcomes.

References

Brouwer, Brigitte, Sophie Waardenburg, Christian Jacobs, Marjori Overdijk, Carsten Leue, Albère Köke, Sander van Kuijk, Maarten van Kleef, Jan Van Zundert, and Nelleke de Meij. 2020. “Biopsychosocial Baseline Values of 15 000 Patients Suffering from Chronic Pain: Dutch DataPain Study.” *Reg. Anesth. Pain Med.* 45 (10): 774–82.

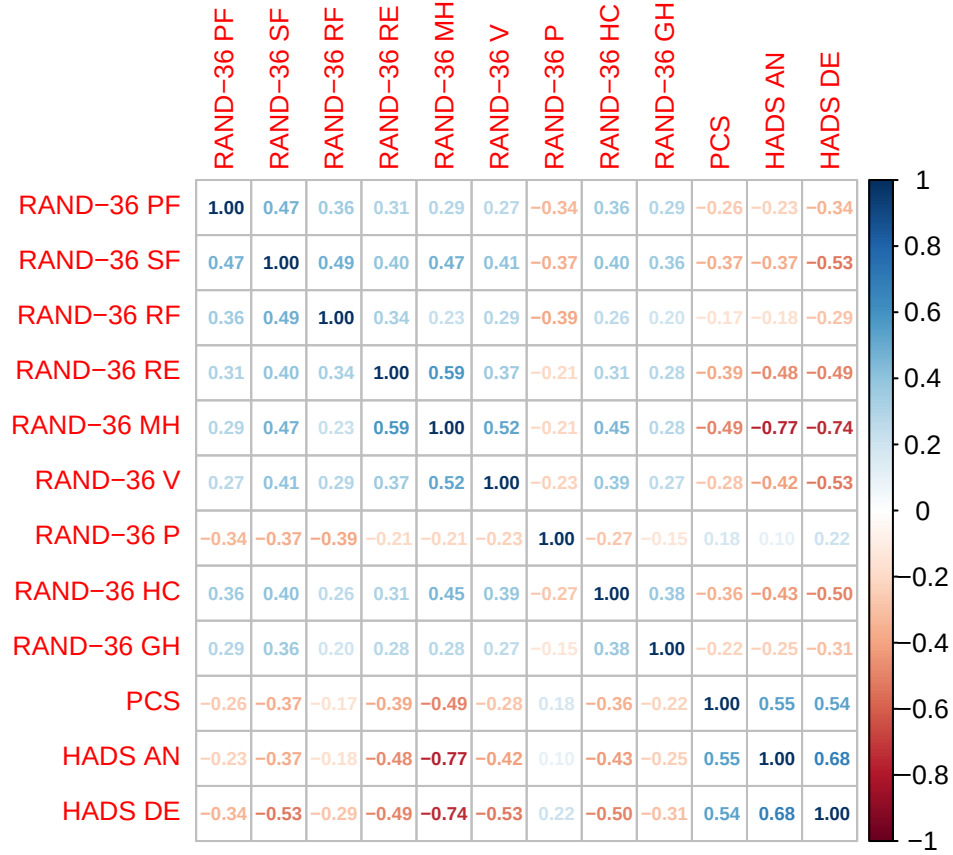


Figure 1: Correlations between patient-reported outcomes. Correlations are expressed as Pearson's correlation coefficient. PF: physical functioning, SF: social functioning, RF: role functioning, RE: role emotional, MH: mental health, V: vitality, P: pain intensity, HC: health change, GH: general health, PCS: pain catastrophizing scale, AN: anxiety, DE: depression.